

Day 27 — Thu 1 Oct

Countdown: 9 days to CS Degree Paper-I

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

🎯 Degree — system software

Source: ComputerScience_Degree/CSD_02_SystemSoftwareAndCompilers.md

Key points — the databases (tables) an assembler keeps

These are the tables you must be able to name and describe — they come up as a 5-marker on their own:

Table	Full name	Contents	Fixed or built?
MOT	Machine Opcode Table	Mnemonic → binary opcode, instruction length, format	Fixed (predefined)
POT	Pseudo-Opcode Table	Assembler directives (START , END , DS , DC , EQU , USING) and the routine that handles each	Fixed
ST	Symbol Table	Each label → its address (value), length, type	Built during Pass 1
LT	Literal Table	Each literal (e.g. =5) → the address assigned to it	Built during Pass 1
Base Table	Base Register Table	Which registers are base registers and their contents (from USING/DROP)	Built
LC	Location Counter	Not a table — a running counter of the current address	Runs in both passes

Common assembler directives (pseudo-ops):

Directive	Meaning
START n	Program begins; set LC = n
END	End of source; may name the entry point
DS n	Define Storage — reserve n units, no initial value
DC v	Define Constant — reserve and initialise with value v
EQU	Equate a symbol to a value (assembly-time constant)
USING/DROP	Declare/undeclare a base register
LTORG	Assemble the literal pool here

2.1 One-pass vs two-pass assemblers

	One-pass	Two-pass

Passes over source	1	2
Forward references	Handled by back-patching (keep a table of unfilled locations, fix them once the symbol is defined)	Resolved naturally: Pass 1 builds the full symbol table before Pass 2 generates code
Speed	Faster (one read)	Slower (two reads)
Memory	Needs a table of pending forward references	Needs the whole symbol table
Simplicity	Trickier logic (patching)	Cleaner, easier to understand and extend
Used when	Load-and-go, memory tight, speed critical	The standard general-purpose design

One-pass idea: when you meet `JMP LOOP` and `LOOP` is undefined, emit the instruction with a blank address and add "location of this blank" to a forward-reference list attached to `LOOP`. When `LOOP` is later defined, walk the list and back-patch every blank with the now-known address.

2.2 The two-pass assembler — what each pass does

Pass 1 — "define the symbols".

- Initialise the Location Counter (LC) from `START`.
- Read each line. For each, look up its length in the MOT/POT and **increment the LC**.
- When a line has a **label**, enter (label → current LC) into the **Symbol Table**.
- Collect **literals** into the **Literal Table** and assign them addresses (at `LTORG` or `END`).
- Process directives that affect the LC (`DS`, `DC`, `EQU`).
- **Output:** a complete Symbol Table + Literal Table, and an intermediate copy of the source with LC values attached. **No machine code is generated yet.**

Pass 2 — "generate the code".

- Re-read the (intermediate) source with the LC known for every line.
- For each instruction: look up the opcode in the MOT, look up every operand symbol in the Symbol Table (now fully populated, so forward references just work), assemble the binary instruction.
- Handle `DC/DS` by emitting/reserving data; place the literal pool.
- **Output:** the object program (machine code + relocation & linking information) and the assembly listing.

Worked two-pass example — trace this by hand

Assume word-addressed memory, each instruction 1 word. Source program:

```

        START 100
        LOAD  A
        ADD   B
        SUB   ONE
        STORE A
        JMP   DONE      ; forward reference — DONE not yet seen
A       DC    5
B       DC    3
ONE     DC    1
DONE    STORE B
        END

```

PASS 1 — assign addresses, build the Symbol Table. LC starts at 100 (from `START 100`).

LC	Label	Statement	Action in Pass 1
100		<code>LOAD A</code>	instruction, len 1 → LC becomes 101

101		ADD B	LC → 102
102		SUB ONE	LC → 103
103		STORE A	LC → 104
104		JMP DONE	forward ref noted; LC → 105
105	A	DC 5	enter A = 105 in ST; LC → 106
106	B	DC 3	enter B = 106 ; LC → 107
107	ONE	DC 1	enter ONE = 107 ; LC → 108
108	DONE	STORE B	enter DONE = 108 ; LC → 109

Symbol Table after Pass 1:

Symbol	Address	Type
A	105	data
B	106	data
ONE	107	data
DONE	108	instruction (label)

PASS 2 — generate object code. Suppose the MOT gives opcodes: LOAD = 01, ADD = 02, SUB = 03, STORE = 04, JMP = 05. Every operand address now comes straight from the completed Symbol Table — the forward reference to **DONE** (address 108) is no longer a problem.

LC	Statement	Opcode	Operand addr	Object code
100	LOAD A	01	105	01 105
101	ADD B	02	106	02 106
102	SUB ONE	03	107	03 107
103	STORE A	04	105	04 105
104	JMP DONE	05	108	05 108 ← forward ref resolved
105	DC 5	—	—	00005 (constant)
106	DC 3	—	—	00003
107	DC 1	—	—	00001
108	STORE B	04	106	04 106

This is the answer an examiner wants for a 10/15-mark "explain the working of a two-pass assembler with an example" question: the two-column pass 1 (address assignment + symbol table) followed by the pass 2 code generation table, with the forward reference explicitly pointed out.

Key points — how a macro processor works

Four data structures / functions:

Component	Role

MDT (Macro Definition Table)	Stores the body of each defined macro
MNT (Macro Name Table)	Maps macro name → its entry (index) in the MDT
ALA (Argument List Array)	Maps formal parameters → actual arguments during an expansion
MDTP / MDI	Macro Definition Table Pointer / index used while expanding

Two logical stages: **(1) macro definition processing** — recognise `MACRO ... MEND`, store the body in the MDT, record the name in the MNT; **(2) macro call expansion** — on a call, look up the name in the MNT, fetch the body from the MDT, substitute the actual arguments (via the ALA) for the formals, and emit the resulting text into the source stream that the assembler then sees.

Worked example — definition and expansion

Definition:

```
MACRO
INCR  &X, &Y          ; &X, &Y are formal (dummy) parameters
LOAD  &X
ADD   &Y
STORE &X
MEND
```

Call: `INCR A, B`

Expansion (the macro processor substitutes `&X→A`, `&Y→B` and copies the body inline):

```
LOAD  A
ADD   B
STORE A
```

A second call `INCR P, Q` expands to a *second, independent copy* with P and Q. That copying is why macros grow code size.

3.1 Nested macros

A macro whose body **calls another macro** (or itself). The processor expands the outer call, and each inner macro call encountered during expansion is itself expanded.

```
MACRO
ADD2  &A, &B
LOAD  &A
ADD   &B
MEND

MACRO
SUM3  &X, &Y, &Z
ADD2  &X, &Y      ; nested call
ADD   &Z
MEND
```

`SUM3 P, Q, R` first expands to `ADD2 P, Q + ADD R`, and `ADD2 P, Q` further expands to `LOAD P + ADD Q`. Final: `LOAD P / ADD Q / ADD R`.

- **Nested definition:** a macro defined *inside* another (the inner one is only defined once the outer is first called).
- **Nested call / recursive macro:** a macro that calls another or itself — needs a stack in the processor and a termination condition (via conditional expansion) to avoid infinite expansion.

3.2 Conditional macro expansion

The body can decide *what* to expand based on the arguments, using **macro-time conditional directives** — **AIF** (assemble if), **AGO** (unconditional branch), **SET** symbols (assembly-time variables), and sequence symbols (labels like **.LOOP**). This is what lets one macro generate different code for different calls, and lets recursive macros terminate.

```
MACRO
GEN    &N
LCLA  &CTR                ; local assembly-time counter
&CTR  SETA  1
.LOOP  AIF  (&CTR GT &N).DONE    ; if CTR > N, jump to .DONE
      WORD &CTR
&CTR  SETA  &CTR+1
      AGO  .LOOP
.DONE  MEND
```

GEN 3 generates **WORD 1 / WORD 2 / WORD 3**. The **AIF** is evaluated **at assembly time**, not run time — no conditional appears in the final code.

Conditional assembly vs run-time if: conditional macro expansion (**AIF**) chooses *which code to generate*; a run-time **if** chooses *which generated code to execute*. A guaranteed distinction MCQ.

Key points — the loader types (learn this table cold; MCQ gold)

Loader type	Allocation	Linking	Relocation	Loading	Idea
Compile-and-go (assemble-and-go)	—	—	—	✓	Assembler places code directly in memory and runs it; no object file. Simple, but re-assembles every run and wastes the assembler's memory.
Absolute loader	prog.	prog.	—	✓ (loader)	Assembler does everything; loader just places code at the fixed address the assembler assumed. Simple & fast, but the programmer must fix addresses and manage linking manually.
Relocating (BSS) loader	loader	—	✓	✓	Assembler emits relocation bits/records : loader can place the program anywhere and fix relocatable addresses. Allows independent module assembly.
Direct-linking loader	loader	✓	✓	✓	The general-purpose loader: does all four functions, using ESD (external symbols), TXT (code), RLD (relocation) and END records. Allows multiple independently compiled modules.
Dynamic loading / dynamic linking	run time	run time	run time	run time	Modules/routines are loaded/linked only when first called , at run time. Saves memory (unused routines never loaded); DLLs / shared objects.

Plus the special one:

- **Bootstrap loader:** the tiny program (in ROM/firmware) that runs at power-on and loads the *first* program — the operating-system loader — into memory. It "pulls itself up by its own bootstraps": a small loader loads a bigger loader that loads the OS. This is the origin of the word **"boot"**.

Direct-linking loader — the object record types (worth naming):

Record	Meaning
ESD — External Symbol Dictionary	Symbols this module defines (for others) and symbols it references (from others)
TXT — Text	The actual machine code and data
RLD — Relocation & Linkage Directory	Which locations need adjusting for relocation/linking
END	End of the module; may give the entry point

🕒 Forensic — science fundamentals

Source: ComputerForensic/FOR_01_ForensicScienceFundamentals.md

1.4 Scope of forensic science in India

Scope = the range of questions it can answer + the range of disciplines it uses.

Domain	Typical questions answered
Forensic Physics	Glass fracture, paint, soil, tool marks, restoration of erased numbers, tyre/footwear marks
Forensic Chemistry	Explosives, petroleum products, fire debris, adulteration, trap cases (phenolphthalein)
Toxicology	Poisons, alcohol, drugs in viscera and body fluids
Biology / Serology	Blood, semen, saliva, hair, fibre, wildlife, diatoms (drowning)
DNA	Identification, paternity/maternity, mass-disaster victim identification, sexual assault
Ballistics	Firearms, cartridge cases, bullet-to-weapon matching, range of firing, GSR
Questioned Documents	Handwriting, signatures, forgery, alterations, ink dating, counterfeit currency
Fingerprints (Dactyloscopy)	Individual identification; AFIS/NAFIS database matching
Forensic Medicine	Autopsy, cause/manner/time of death, injuries
Forensic Psychology	Polygraph, narco-analysis, brain electrical oscillation signature (BEOS) — note the <i>Selvi v. State of Karnataka</i> (2010) restriction that these cannot be conducted without consent ⚠ verify citation before quoting
Cyber / Digital Forensics	Computers, mobiles, networks, cloud, CCTV/audio-video authentication
Forensic Anthropology / Odontology	Skeletal remains, age/sex estimation, bite marks, superimposition

Growth drivers in India: rising cybercrime, the 2024 criminal-law overhaul mandating forensic teams for serious offences, NAFIS rollout, expansion of DNA facilities, and NFSU-driven manpower.

Constraints: pendency and backlog in FSLs, shortage of trained examiners, uneven State capacity, delays affecting admissibility.

1.5 Basic principles of forensic science — the core of Unit 1

These six (some books say seven) principles are near-certain exam material. Learn each as **Statement** → **Meaning in plain English** → **Example** → **Digital-forensics analogue**. The digital analogue is a strong differentiator in the answer sheet for *this* elective.

(i) Locard's Exchange Principle — "Every contact leaves a trace"

Statement: When any two objects come into contact, there is always a **mutual exchange of material** between them.

Plain English: You cannot walk into a room without both leaving something behind and taking something away. The criminal brings something to the scene and carries something from it. The exchange may be too small to find, but it always occurs.

Example: A burglar breaking a window leaves fingerprints and fibres on the sill, and carries away glass fragments in his shoe soles and trouser turn-ups.

Digital analogue: A user cannot open a file without leaving traces — an MFT timestamp change, a prefetch entry, an LNK file, a registry MRU entry, a browser cache entry. This is the single most quotable line you can put in a digital-forensics answer.

Corollary / limitation: the trace's *detection* depends on the time elapsed, the medium and the analyst's skill. Absence of trace evidence is not proof of absence of contact.

(ii) Law of Individuality — "Every object, natural or man-made, is unique"

Statement: Every object, natural or artificial, has an individuality which is not duplicated in any other object. No two objects, **not even two grains of sand or two identical twins' fingerprints**, are exactly alike.

Plain English: Manufacturing tolerances, wear, growth and random processes make every item one-of-a-kind at a fine enough level of examination.

Example: Two bullets fired from the *same* barrel bear the same striations; bullets from two "identical" barrels of the same model do not, because the rifling tools wear differently. Identical twins share DNA but **not** fingerprints (ridge detail is formed by random foetal pressures).

Digital analogue: Cryptographic hash values — two different files essentially never produce the same SHA-256. Also MAC addresses, IMEI numbers, hardware serial numbers.

Exception to note in the answer: DNA of monozygotic twins is (practically) identical — the usual exception cited against this law for DNA evidence.

(iii) Principle of Comparison — "Only like can be compared with like"

Statement: Only comparable samples/standards can be compared with the questioned evidence. A questioned sample must be compared against a **control/standard/known** sample of the same type, collected under comparable conditions.

Plain English: You cannot identify a suspect's handwriting by comparing his printed capitals to a cursive ransom note. You must obtain a specimen in the same script, same writing instrument, same posture.

Example: To match a tyre mark you need a test impression from the suspect tyre on similar substrate. To match a bullet you fire test bullets from the seized weapon into a recovery tank.

Digital analogue: Compare the hash of the *working copy* to the hash of the *original* image; compare a carved JPEG against a known-good exemplar from the same camera model.

(iv) Principle of Analysis — "The analysis is only as good as the sample"

Statement: The result of an analysis can be no better than the **sample analysed**, the **method used** and the **person who performs** it. Incorrect sampling, contaminated sampling or an unqualified analyst destroys the value of even a perfect instrument.

Plain English: Garbage in, garbage out. This is the principle that makes crime-scene collection technique legally important.

Example: A blood sample collected into an unsterile container and left in a hot vehicle yields a useless DNA result no matter how good the lab is.

Digital analogue: Imaging a live disk without a write blocker corrupts timestamps; the subsequent analysis is worthless. Also: analysis of a *logical* copy cannot recover deleted files, because the sample never contained them.

(v) Law of Progressive Change — "Everything changes with the passage of time"

Statement: Everything changes with time. The rate of change varies with the object and the conditions. The criminal changes (appearance, memory), the scene changes (weather, traffic, contamination), and the evidence changes (decomposition, evaporation, corrosion).

Plain English: Evidence has a shelf life. Delay is the enemy.

Example: A body decomposes; a bloodstain degrades; a footprint in mud dries and cracks; volatile accelerants evaporate from fire debris within hours.

Digital analogue: This is precisely why the **order of volatility** exists — RAM contents vanish on power-off, ARP caches age out in minutes, DHCP leases and ISP logs are overwritten in days or weeks. Cite this link in the exam; it directly connects TP-I §1 to TP-II §1.

(vi) Principle of Circumstantial Facts / "Facts do not lie, men can and do"

Statement: Physical/circumstantial evidence does not lie, cannot be wholly absent, and is not subject to the frailties of human memory or motive — but it **must be interpreted correctly**. Oral testimony can be false, mistaken, coerced or retracted; physical evidence cannot.

Plain English: A witness may turn hostile in court; a fingerprint will not.

Example: In a case where all eyewitnesses turn hostile, the recovered weapon bearing the accused's fingerprints and the victim's blood can still sustain a conviction.

Digital analogue: Log files and hash values do not change their story on cross-examination — provided integrity is proved (§65B / BSA §63 certificate).

Caveat: physical evidence is only as truthful as its *interpretation*; wrongful convictions arise from over-claimed forensic conclusions (bite marks, hair microscopy, arson "indicators").

Some texts add a seventh: **Principle of Presentation** — the finding must be presented in a form the court can understand. Mention it as an add-on, not as a core principle.

Memory hook

L-I-C-A-P-C: Locard, Individuality, Comparison, Analysis, Progressive change, Circumstantial facts.

Or a sentence: "Little Items Can Always Prove Crime."

1.6 Significance of forensic science

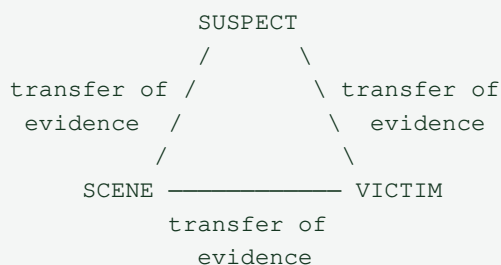
1. **Objective, scientific proof** independent of witness reliability.
2. **Links** the four corners of a crime: the **victim**, the **suspect**, the **scene**, and the

weapon/instrument — the "forensic linkage triangle/quadrangle" (draw it, see below).

3. **Excludes the innocent** as often as it convicts the guilty (exculpatory value; DNA exonerations).
4. **Reconstructs** the sequence of events (bloodstain pattern analysis, trajectory analysis, log timeline reconstruction).
5. **Identifies** persons and objects (fingerprints, DNA, IMEI, hashes).
6. **Deterrent effect** and increased conviction rate.
7. **Corroborates or contradicts** the confession/testimony on record.
8. Handles crimes where there **are no witnesses at all** — the normal case in cybercrime.

Limitations (add these for a 10-mark answer): backlog and delay; contamination risk; dependence on the quality of scene work; interpretive/expert bias; cost and infrastructure; absence of statutory forensic standards in some domains; over-reliance leading to the "CSI effect" in courts.

Diagram to hand-draw: the Linkage Triangle



(Weapon/instrument sits in the middle, connecting all three)

Draw three labelled circles — SUSPECT, VICTIM, SCENE — joined by double-headed arrows labelled "exchange of trace material (Locard)". Put WEAPON in the centre with lines to all three. One extra mark, every time.

2.3 Directorate of Forensic Science Services (DFSS)

Attribute	Detail
Parent	Ministry of Home Affairs, Government of India
Created	2002 (as Directorate of Forensic Science, carved out of BPR&D); later renamed DFSS
Headquarters	New Delhi
Head	Director-cum-Chief Forensic Scientist
Controls	The CFSLs and the GEQD offices

Functions of DFSS

1. Administrative and technical control of CFSLs and GEQDs.
2. Provides forensic services to **central investigating agencies** (CBI, NIA, ED, NCB, CAPFs) and to State police in referred/sensational/inter-State cases.
3. **Referral laboratory** function — second opinion where a State FSL report is disputed.
4. Formulates **standards, protocols, SOPs and quality-assurance** norms for the country.
5. Research & development in forensic techniques.
6. **Training** of forensic scientists, police officers, judicial officers and prosecutors.
7. Advises MHA on forensic policy and on new legislation.

8. Coordinates the national forensic infrastructure and manpower planning.

2.4 Central Forensic Science Laboratories (CFSLs)

Full-fledged multidisciplinary central laboratories. The classical list (△ verify the current count — the network has been expanded in recent years and new CFSLs/NFSU campuses have been sanctioned):

CFSL	Notable speciality / note
Kolkata (1957)	The oldest CFSL. Historically the centre for Questioned Documents / handwriting ; also serology, ballistics
Hyderabad	Chemistry, toxicology, narcotics , DNA; serves southern region
Chandigarh	Broad multidisciplinary; serves northern region
Guwahati	Serves the North-East , including Nagaland — <i>know this one, it is the regionally relevant answer</i>
Bhopal	Multidisciplinary (central region)
Pune	Multidisciplinary (western region)
New Delhi (CFSL, CBI)	Under the administrative control of the CBI , not DFSS — the standard exam trick. Strong in DNA, cyber, documents

Under the National Forensic Infrastructure Enhancement Scheme (NFIES) further CFSLs/NFSU campuses have been sanctioned across States △ verify current status and list.

Functions of a CFSL

1. Examination of exhibits referred by central agencies, courts, and State police in important cases.
2. Referral/second-opinion analysis.
3. Crime-scene attendance in sensational cases.
4. Expert testimony in court by the examining scientist.
5. R&D and method validation.
6. Training and internships.

2.6 State Forensic Science Laboratories (SFSL)

Attribute	Detail
Control	State Home Department / State Police (Police is a State subject)
Location	Normally the State capital; one per State/UT
Head	Director, SFSL
Users	State police, State courts, State agencies

Functions: routine casework for the entire State; crime-scene visits in serious cases; expert testimony in State courts; supervising the RFSLs and mobile units; training State police in evidence collection; maintaining State-level databases.

Nagaland-relevant note: the North-East is served centrally by **CFSL Guwahati**; State-level work in Nagaland is handled by the State FSL under the Nagaland Police △ verify current status and location before writing it as a fact.

2.9 Divisions inside a full-service forensic science laboratory

This is a very common 5- or 10-mark question: "Describe the various divisions of a Forensic Science Laboratory and their functions." Answer as a table.

Division	What it examines / does
Biology	Hair, fibre, wood, pollen, diatoms (drowning), botanical material, wildlife material
Serology	Blood, semen, saliva, sweat, urine; species of origin; blood grouping; stain identification
DNA / Molecular Biology	STR profiling, paternity/maternity, sexual-assault cases, disaster victim identification, DNA databanking
Chemistry	Explosives and post-blast residue, petroleum products, fire debris/accelerants, dyes, adulterants, trap-case (phenolphthalein) analysis, general unknowns
Toxicology	Poisons, pesticides, alcohol, drugs of abuse in viscera, blood and body fluids
Narcotics / NDPS	Identification and quantification of narcotic drugs and psychotropic substances
Physics	Glass, paint, soil, cement, tool marks, tyre and footwear marks, restoration of erased serial numbers, accident reconstruction
Ballistics	Firearms, ammunition, cartridge cases, bullets, range and direction of fire, gunshot residue (GSR)
Documents / Questioned Documents	Handwriting, signatures, forgery, alterations, erasures, indented writing, ink and paper analysis, counterfeit currency, typewriting/printing
Fingerprint / Dactyloscopy	Development, lifting, comparison and identification of latent prints; AFIS/ NAFIS searching (usually under the Finger Print Bureau)
Cyber Forensics / Computer Forensics	Computers, hard disks, mobile phones, memory cards, network and cloud data, CCTV, audio-video authentication, malware analysis
Photography / Imaging	Crime-scene and exhibit photography, photomicrography, infrared/UV imaging, image enhancement
Lie Detection / Forensic Psychology	Polygraph, narco-analysis, BEOS — subject to consent requirements
Prohibition & Excise	Liquor, illicit spirits, methanol
Anthropology / Odontology (where present)	Skeletal remains, age/sex/stature estimation, superimposition, bite marks